

A Technical Introduction to Gated recurrent unit

Section 1

Architecture There are several variations on the full gated unit, with gating done using the previous hidden state and the bias in various combinations, and a simplified form called minimal gated unit. In the following, the operator $\odot$ denotes the Hadamard product.

Section 2

$$\begin{aligned} z_t &= \sigma(W_z x_t + U_z h_{t-1} + b_z) \\ r_t &= \sigma(W_r x_t + U_r h_{t-1} + b_r) \\ \tilde{h}_t &= \tanh(W_h x_t + U_h r_t \odot h_{t-1} + b_h) \\ h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \end{aligned}$$

Section 3

$W \in \mathbb{R}^{e \times d}$, $U \in \mathbb{R}^{e \times e}$ and $b \in \mathbb{R}^e$: parameter matrices and vector which need to be learned during training

Activation functions

Section 4

$$\begin{aligned} z_t &= \sigma(BN(W_z x_t + U_z h_{t-1} + b_z)) \\ \tilde{h}_t &= \tanh(BN(W_h x_t + U_h h_{t-1} + b_h)) \\ h_t &= z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \end{aligned}$$

Section 5

ϕ : The original is a hyperbolic tangent. Alternative activation functions are possible, provided that $\sigma(x) \in [0, 1]$.

Section 6

Minimal gated unit The minimal gated unit (MGU) is similar to the fully gated unit, except the update and reset gate vector is merged into a forget gate. This also implies that the equation for the output vector must be changed:

Section 7

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f) \\ \tilde{h}_t &= \tanh(W_h x_t + U_h (f_t \odot h_{t-1} + b_h)) \\ h_t &= (1 - f_t) \odot h_{t-1} + f_t \odot \tilde{h}_t \end{aligned}$$

$$(W_h x_t + U_h (f_t \odot h_{t-1} + b_h)) \odot h_{t-1} + f_t \odot \tilde{h}_t$$

Section 8

$$\begin{aligned} z_t &= \sigma(U_z h_{t-1} + b_z) \\ r_t &= \sigma(U_r h_{t-1} + b_r) \\ \tilde{h}_t &= \tanh(U_h h_{t-1} + b_h) \\ h_t &= z_t \odot h_{t-1} + r_t \odot \tilde{h}_t \end{aligned}$$

Section 9

LiGRU has been studied from a Bayesian perspective. This analysis yielded a variant called light Bayesian recurrent unit (LiBRU), which showed slight improvements over the LiGRU on speech recognition tasks.

Section 10

In artificial neural networks, the gated recurrent unit (GRU) is a gating mechanism used in recurrent neural networks, introduced in 2014 by Kyunghyun Cho et al. The GRU is like a long short-term memory (LSTM) with a gating mechanism to input or forget certain features, but lacks a context vector or output gate, resulting in fewer parameters than LSTM. GRU's performance on certain tasks of polyphonic music modeling, speech signal modeling and natural language processing was found to be similar to that of LSTM. GRUs showed that gating is indeed helpful in general, and Bengio's team came to no concrete conclusion on which of the two gating units was better.